GBD · PROBLEM SET 1 · SOLUTIONS

Fundamental concepts of graphs

Step-by-step solutions to the 11 exercises. USTHB, M1 TGBD (Graph Theory for Big Data), 2026/2027. Course: *Concepts fondamentaux*.

Tools used in this sheet

Handshake lemma
Σ d(x) = 2m, so the number of odd-degree vertices is even
Ex 1, 2, 6, 7

Havel–Hakimi
remove the largest degree d, subtract 1 from the next d terms, repeat
decides if a sequence is graphical (Ex 1)

Euler
connected + all degrees even ⇔ Eulerian cycle; exactly 2 odd ⇒ Eulerian chain
Ex 4, 6, 10

Bipartite
bipartite ⇔ no odd cycle (course §1.1)
Ex 2, 4, 7

Isomorphism invariants
order, size, degree sequence, cycles, induced subgraphs on equal degrees
a different invariant proves non-isomorphism (Ex 3)

Property 1 (course)
S stable ⇔ V \ S transversal; max|S| + min|T| = n
Ex 11

Vocabulary follows the course: chain (chaîne), elementary chain (chaîne élémentaire, no repeated vertex), cycle, stable set (stable), transversal. Every number in this sheet (degree sums, isomorphisms, matrix powers, colourings, τ and α) was checked by a Python program (networkx).

Exercise 1 Degrees, handshakes and graphical sequences

ASKED
I) 20 people with 7×3, 9×4, 4×5 friends? II) Handshakes at a meeting of 4 couples: possible values, and Mr X's count. III) Which sequences are degree sequences of a simple graph?

METHOD
Handshake lemma Σ d(x) = 2m (the sum of degrees is even). The degree of a vertex in a simple graph of order n is at most n − 1. **Havel–Hakimi** to build or refute a sequence.

1 I Sum of the degrees

Model: vertices = people, edges = friendships. Then

$$\Sigma\, d(x) = 7\cdot 3 + 9\cdot 4 + 4\cdot 5 = 21 + 36 + 20 = \mathbf{77}$$

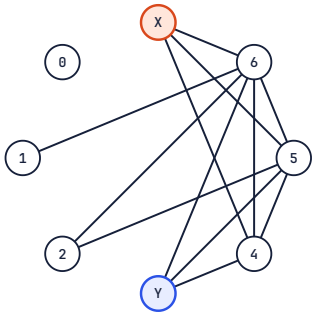
77 is odd, but Σ d(x) = 2m is always even. **Impossible.** (Equivalently: the 7 + 4 = 11 people of odd degree would be an odd number of odd-degree vertices.)

2 II Possible values

Graph: 8 vertices (4 couples), edge = handshake. A person cannot shake their own hand or their spouse's, so each degree is in {0, 1, ..., 6}: **7 possible values**. The 7 people other than Mr X have 7 distinct degrees, so they take **exactly the values 0, 1, 2, 3, 4, 5, 6**.

3 II Pair the spouses

- The person with 6 shook hands with everyone except their spouse. Everyone except that spouse therefore has degree ≥ 1, so the spouse is the person with **0**: couple (6, 0).
- Remove this couple: every remaining person loses exactly 1 handshake (the one with "6"). The degrees become 0...4 plus Mr X, and the same argument pairs **(5, 1)**, then **(4, 2)**.
- The only person left among the seven is the one with **3**: this is Mrs X. Mr X shook hands with 6, 5, 4 and not with 2, 1, 0 (they had used up their counts), so Mr X has **3**.



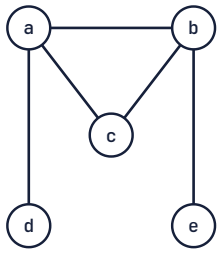
`one valid configuration: label = handshakes; X = Mr X, Y = Mrs X`

Check: a program tried all 2^{24} handshake patterns; the 48 that satisfy the statement all give Mr X = 3 and Mrs X = 3.

4 III Test each sequence

Sequence	n	Σ	Verdict
7, 6, 5, 4, 3, 2, 1	7	28	No: degree 7 > n - 1 = 6
6, 6, 5, 4, 3, 3, 1	7	28	No (see below)
3, 3, 2, 1, 1	5	10	Yes

- Sequence 2: the two vertices of degree 6 = n − 1 are adjacent to every other vertex, so every vertex has degree ≥ 2. A vertex of degree 1 cannot exist.
- Sequence 3, Havel–Hakimi: (3,3,2,1,1) → remove 3: (2,1,0,1) → sort (2,1,1,0) → remove 2: (0,0,0) → **graphical**. A realisation:



degrees a,b = 3; c = 2; d,e = 1

ANSWER

I) Impossible (degree sum 77 is odd). II) Values 0…6; spouses pair as (6,0), (5,1), (4,2); Mrs X has 3 and **Mr X shook 3 hands**. III) 1 no, 2 no, 3 yes.

Exercise 2 Five general properties

ASKED
For a graph G of order n and size m, prove the five statements.

METHOD
Handshake lemma, counting edges between the two sides of a bipartition, the pigeonhole principle, and the handshake lemma applied to one connected component.

1 $m \le n(n-1)/2$ and $\delta \le 2m/n \le \Delta$

An edge is a pair of distinct vertices, and there are $C(n,2) = n(n-1)/2$ pairs: $m \le n(n-1)/2$ (equality for K_n).

Each degree satisfies $\delta \le d(x) \le \Delta$. Summing over the n vertices: $n\delta \le \Sigma\, d(x) = 2m \le n\Delta$, and dividing by n gives **$\delta \le 2m/n \le \Delta$** ($2m/n$ is the average degree).

2 Bipartite ⇒ $m \le n^2/4$

Let $V = V_1 \cup V_2$ with $|V_1| = p$, $|V_2| = n - p$. Every edge joins V_1 to V_2, so $m \le p(n-p)$. And

$$p(n-p) = n^2/4 - (p - n/2)^2 \le n^2/4$$

3 Regular with n odd ⇒ not bipartite

Let G be r-regular with $r \ge 1$ and suppose G bipartite with sides V_1, V_2. Count the edges from each side: every edge has exactly one end in V_1, so $m = r|V_1|$, and likewise $m = r|V_2|$. Hence $|V_1| = |V_2|$ and $n = 2|V_1|$ is even. Contradiction.

$r \ge 1$ is needed: the edgeless graph on 3 vertices is 0-regular and bipartite.

4 Two vertices with the same degree ($n \ge 2$)

Degrees lie in $\{0, \ldots, n-1\}$: n values for n vertices. But 0 and $n-1$ cannot both occur: a vertex of degree $n-1$ is adjacent to all others, so no vertex has degree 0. So the n degrees take at most $n-1$ values, and by the pigeonhole principle two vertices share a degree.

5 Exactly two odd vertices x, y ⇒ a chain links them

Let C be the connected component of x. The subgraph C is a graph in its own right, so by the handshake lemma it has an even number of odd-degree vertices (degrees in C are the same as in G). x is one of them, so C contains another odd vertex, and the only one is y. So x and y are in the same component: there is a chain from x to y.

ANSWER
1) count pairs, then sum $\delta \le d(x) \le \Delta$. 2) $m \le p(n-p) \le n^2/4$. 3) $r|V_1| = r|V_2|$ forces n even. 4) pigeonhole on $n-1$ values. 5) handshake lemma inside the component of x.

Exercise 3 — Isomorphism

ASKED

I) Are G_1 and H_1, and G_2 and H_2, isomorphic? II) Induced subgraphs on vertices of degree r: prove invariance, and give a counter-example to the converse.

METHOD

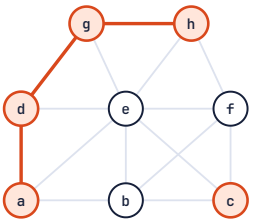
To prove **non-isomorphism**, find an invariant that differs. To prove **isomorphism**, give an explicit bijection f and check that it maps every edge to an edge (same number of edges, so non-edges follow).

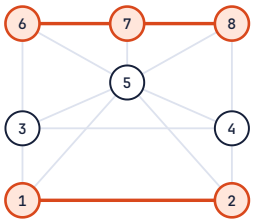
1 I Compare the simple invariants of G_1, H_1

	n	m	degree sequence
G_1	8	15	7, 4, 4, 3, 3, 3, 3, 3
H_1	8	15	7, 4, 4, 3, 3, 3, 3, 3

G_1: d(e) = 7, d(b) = d(f) = 4, the others 3. H_1: d(5) = 7, d(3) = d(4) = 4, the others 3. All equal, so we look deeper.

2 I Look at the vertices of degree 3





G_1: degree-3 vertices induce $P_4 + K_1$ H_1: degree-3 vertices induce $P_3 + P_2$

- G_1: S = {a, c, d, g, h}, edges ad, dg, gh: a chain a–d–g–h of length 3 plus the isolated vertex c.
- H_1: T = {1, 2, 6, 7, 8}, edges 12, 67, 78: a chain 6–7–8 of length 2 plus the edge 1–2.

An isomorphism preserves degrees, so it would map S onto T and $G_1(S)$ onto $H_1(T)$ (part II.1). These two subgraphs are not isomorphic (one has an isolated vertex, the other none), so $G_1 \ncong H_1$.

3 I G_2 and H_2

Both have n = 10, m = 15, are 3-regular, have no triangle and no 4-cycle (shortest cycle length 5), diameter 2: both are drawings of the **Petersen graph**. An explicit isomorphism:

```
a→2   b→3   c→4   d→5   e→9   f→1   g→8   h→10   i→7   j→6
```

Check (a few edges): fe → 1 9, fa → 1 2, fh → 1 10, ij → 7 6, bc → 3 4; all 15 edges of G_2 map to edges of H_2 (verified by program). $G_2 \cong H_2$.

4 II.1 Isomorphic ⇒ G(S) ≅ H(T)

Let f : V(G) → V(H) be an isomorphism. f preserves adjacency both ways, so N(f(x)) = f(N(x)) and $d_H(f(x)) = d_G(x)$. Hence f(S) = T. The restriction $f|_S : S \to T$ is a bijection, and for x, y ∈ S, xy ∈ E(G(S)) ⇔ xy ∈ E(G) ⇔ f(x)f(y) ∈ E(H) ⇔ f(x)f(y) ∈ E(H(T)). So $f|_S$ is an isomorphism G(S) → H(T).

5 II.2 Counter-example

G_1 and H_1 above with r = 3: same order (8), same size (15), same degree sequence, but $G_1(S) = P_4 + K_1$ and $H_1(T) = P_3 + P_2$ are not isomorphic. (Smaller example: C_6 and two disjoint triangles, r = 2, where S = T = V.)

ANSWER

$G_1 \ncong H_1$ (degree-3 vertices induce different subgraphs). $G_2 \cong H_2$ (both are the Petersen graph; bijection above). II.2: G_1, H_1 with r = 3.

Exercise 4 The hypercube Q_n

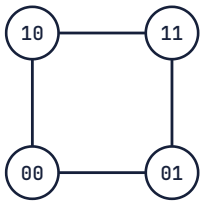
ASKED
Draw Q_1, Q_2, Q_3; order, degrees, size; bipartite; Eulerian; Q_n as a Cartesian product and Hamiltonicity; distance and diameter.

METHOD
Vertices are binary words of length n; adjacent ⇔ Hamming distance 1 (course §1.4). Weight w(x) = number of 1s.

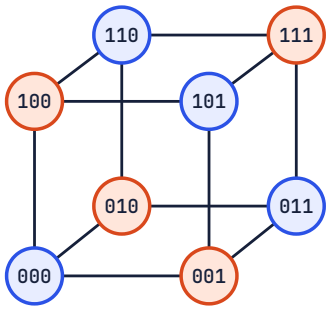
1 Drawings, order, degree, size



$Q_1 = K_2$



$Q_2 = C_4$



Q_3: blue = even weight, orange = odd

- **Order**: $|\{0,1\}^n| = \mathbf{2^n}$.
- **Degree**: a neighbour of x is obtained by flipping one of the n coordinates, so every vertex has degree **n** (Q_n is n-regular).
- **Size**: $2m = \Sigma\, d(x) = n \cdot 2^n$, so $\mathbf{m = n \cdot 2^{n-1}}$. ($Q_3$: 3·4 = 12.)

2 Q_n is bipartite

V_1 = words of even weight, V_2 = words of odd weight. Flipping one bit changes the weight by ±1, so every edge joins V_1 to V_2, and both sets are stable. Hence Q_n is bipartite (colours in the figure).

3 Eulerian

Q_n is connected (flip the differing bits one at a time) and every degree equals n. By Euler: **Q_n is Eulerian ⇔ n is even**.

4 Cartesian product and Hamiltonian cycle

$Q_n = K_2 \,\square\, Q_{n-1} = \square^n K_2$: the first bit is a vertex of K_2, the other n − 1 bits a vertex of Q_{n-1}, and two words are adjacent iff they differ in the first bit only (edge of K_2, same rest) or in the rest only.

Induction. $Q_2 = C_4$ is Hamiltonian. If $x_1, \ldots, x_N$ ($N = 2^{n-1}$) is a Hamiltonian cycle of Q_{n-1}, then

$$0x_1,\ 0x_2,\ \ldots,\ 0x_N,\ 1x_N,\ 1x_{N-1},\ \ldots,\ 1x_1,\ \text{(back to } 0x_1\text{)}$$

is a Hamiltonian cycle of Q_n: each step changes one bit, and $0x_N$–$1x_N$ and $1x_1$–$0x_1$ are edges of the K_2 copy. For n = 3: 000, 001, 011, 010, 110, 111, 101, 100 (the Gray code).

5 Distance and diameter

Let h(x, y) = number of coordinates where x and y differ. Each edge changes one coordinate, so any chain from x to y has length ≥ h(x, y). Flipping the differing coordinates one by one gives a chain of length exactly h(x, y). So **d(x, y) = h(x, y)**, the Hamming distance. The maximum is reached for complementary words (e.g. 0…0 and 1…1): **diam(Q_n) = n**.

	order	degree	size	Eulerian	diam
Q_1	2	1	1	no	1
Q_2	4	2	4	yes	2
Q_3	8	3	12	no	3
Q_4	16	4	32	yes	4
Q_5	32	5	80	no	5
Q_6	64	6	192	yes	6

Table checked by program, including d = Hamming distance for every pair, n ≤ 6.

ANSWER

Order 2^n, n-regular, size $n \cdot 2^{n-1}$; bipartite by parity of weight; Eulerian ⇔ n even; $Q_n = K_2 \square Q_{n-1}$, Hamiltonian for n ≥ 2 (Gray code); d(x, y) = Hamming distance, diameter n.

Exercise 5 Three colours on a connected graph

ASKED

G connected, n ≥ 3, every vertex red, blue or green, each colour used. 1) Two adjacent vertices of different colours. 2) It can happen that no two adjacent vertices share a colour. 3) A chain meets all three colours.

METHOD

Walk along a chain between two vertices of different colours; for 3, attach a shortest chain from the third colour to the first chain.

1

1 Adjacent vertices of different colours

Take x red and y blue. G is connected, so there is a chain $x = u_0, u_1, \ldots, u_k = y$. If every edge u_iu_{i+1} joined vertices of the same colour, all u_i would have the colour of x, but y is blue. So some edge u_iu_{i+1} joins two differently coloured vertices.

2

2 No monochromatic edge, for every n ≥ 3

Take the path $P_n = (1, 2, \ldots, n)$ and colour vertex i with colour i mod 3 (red, blue, green, red, ...). Consecutive vertices always get different colours, all three colours appear because n ≥ 3, and this works for every n. (A star $K_{1,n-1}$ with a red centre and blue and green leaves also works.)



P_n coloured red, blue, green, red, … (here n = 6)

3

3 A chain with all three colours

Take x red, y blue, z green, and an elementary chain P from x to y. If P contains a green vertex, done. Otherwise, take a shortest chain Q from z to a vertex of P; it meets P only at its last vertex w. w is red or blue (P has no green):

- w blue: x →(P) w →(Q) z has red x, blue w, green z.
- w red: y →(P) w →(Q) z has blue y, red w, green z.

Both are elementary chains (Q avoids P except at w), so the chain exists.

ANSWER

1) Along a chain from a red to a blue vertex the colour must change on some edge. 2) Path coloured red, blue, green periodically. 3) Join the green vertex to a red–blue chain at w and keep the half of the chain whose far end has the colour w lacks.

Exercise 6 The line graph L(G)

ASKED

1) L(G) has m vertices and Σ C(d(x), 2) edges. 2) G Eulerian ⇒ L(G) Hamiltonian and Eulerian. 3) The complement of L(K_5) is the Petersen graph P(5, 2).

METHOD

Vertices of L(G) = edges of G; two are adjacent iff they share an end. Count adjacent pairs vertex by vertex. Read an Eulerian cycle of G as a sequence of edges.

1 Order and size

By definition V(L(G)) = E(G): **m vertices**. Two edges of a simple graph share at most one end. At a vertex x, the d(x) incident edges are pairwise adjacent, giving C(d(x), 2) pairs, and each adjacent pair is counted at exactly one vertex (their common end). So

$$|E(L(G))| = \Sigma_{x \in V}\, C(d(x), 2) = \Sigma\, d(x)(d(x) - 1)/2$$

Check: K_5 has m = 10 and 5·C(4,2) = 30 edges in L(K_5); Petersen: m = 15, 10·C(3,2) = 30.

2 G Eulerian ⇒ L(G) Hamiltonian and Eulerian

Let (e_1, e_2, …, e_m) be the edges in the order of an Eulerian cycle of G. Consecutive edges share a vertex, and so do e_m and e_1 (the cycle is closed). Every edge appears exactly once, so e_1, …, e_m, e_1 is a **Hamiltonian cycle of L(G)** (m ≥ 3 since a simple Eulerian graph contains a cycle).

L(G) is connected (it has a Hamiltonian cycle). The edge xy has degree d(x) − 1 + d(y) − 1 = d(x) + d(y) − 2 in L(G), which is even since all degrees of G are even. So **L(G) is Eulerian**.

3 Complement of L(K_5)

Vertices of L(K_5): the 10 pairs {i, j} ⊂ {1, …, 5}. In L(K_5) two pairs are adjacent iff they **intersect**, so in the complement they are adjacent iff they are **disjoint**. This is the Kneser graph K(5,2), the classic definition of the Petersen graph:

- order 10; each pair {i, j} is disjoint from C(3,2) = 3 pairs: 3-regular, size 15 (= 45 − 30);
- outer 5-cycle 12–34–51–23–45–12, inner pentagram 35–24–13–25–14–35, spokes 12–35, 34–25, 51–24, 23–14, 45–13: exactly the drawing of P(5, 2).

Isomorphism with networkx's Petersen graph verified by program.

ANSWER

1) m vertices, Σ C(d(x),2) edges. 2) An Eulerian cycle lists the edges as a Hamiltonian cycle of L(G); degrees d(x) + d(y) − 2 are even. 3) Complement of L(K_5) = "disjoint pairs" graph = Petersen.

Exercise 7 Complete multipartite graphs, Petersen, forests

ASKED
Size of $K_{n_1,\dots,n_k}$ and its complement; Petersen numbers; when $K_{p,q}$ is regular; max size of a bipartite graph; size of a forest.

METHOD
Count edges as pairs of vertices in different parts; complement = remove those; maximise $p(n-p)$ over integers.

1 Size of $K_{n_1,\dots,n_k}$

Every vertex is adjacent exactly to the vertices outside its own part, so $d(x) = n - n_i$ for $x \in V_i$. Then

$$2m = \Sigma_i\, n_i(n - n_i) = n\cdot\Sigma\, n_i - \Sigma\, n_i^2 = n^2 - \Sigma\, n_i^2 \Rightarrow m = \tfrac{1}{2}(n^2 - \Sigma\, n_i^2)$$

$K_{3,2,2}$: $n = 7$, $m = \tfrac{1}{2}(49 - 9 - 4 - 4) = \mathbf{16}$. Direct count: $3\cdot2 + 3\cdot2 + 2\cdot2 = 6 + 6 + 4 = 16$ ✓.

2 Complement

Inside a part there are no edges, between parts all edges. In the complement it is the opposite: **complement of $K_{n_1,\dots,n_k} = K_{n_1} \cup K_{n_2} \cup \dots \cup K_{n_k}$** (disjoint cliques). Check sizes: $\Sigma\, C(n_i, 2) + m = C(n, 2)$. For $K_{3,2,2}$: $3 + 1 + 1 = 5$ and $5 + 16 = 21 = C(7, 2)$ ✓.

3 Petersen graph and its complement

$m = 10\cdot3/2 = \mathbf{15}$. The complement has order **10**, size $C(10, 2) - 15 = 45 - 15 = \mathbf{30}$, and is **6-regular** (each vertex: $9 - 3 = 6$).

4 $K_{p,q}$ regular $\Leftrightarrow$ $p = q$

In $K_{p,q}$ ($p, q \geq 1$) the p vertices of the first side have degree q and the q others have degree p. It is regular iff $p = q$. An r-regular complete bipartite graph is therefore $K_{r,r}$: **order $2r$, size r^2**.

5 $m \leq \lfloor n^2/4 \rfloor$

With sides of sizes p and $n - p$, $m \leq p(n - p)$ (Ex 2.2). m and $p(n - p)$ are integers, and $p(n - p)$ is largest when $p = \lfloor n/2 \rfloor$, giving $\lfloor n/2 \rfloor\cdot\lceil n/2 \rceil = \lfloor n^2/4 \rfloor$ ($n = 2k$: k^2; $n = 2k + 1$: $k(k + 1) = \lfloor (4k^2 + 4k + 1)/4 \rfloor$). **Equality** needs every V_1–V_2 pair to be an edge and $|p - (n - p)| \leq 1$: the graph is $\mathbf{K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}}$.

6 Forest with c components

The components $T_1, \dots, T_c$ are trees of orders $p_1, \dots, p_c$ with $\Sigma\, p_i = n$. Tree T_i has $p_i - 1$ edges, so $m = \Sigma\, (p_i - 1) = \mathbf{n - c}$.

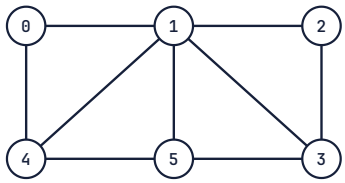
ANSWER
1) $m = \tfrac{1}{2}(n^2 - \Sigma\, n_i^2)$; $K_{3,2,2}$: 16. 2) Disjoint union of K_{n_i}. 3) Petersen: 15 edges; complement: order 10, size 30, 6-regular. 4) $K_{r,r}$: order $2r$, size r^2. 5) Equality only for $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$. 6) $n - c$ edges.

Exercise 8 Matrices and adjacency lists

ASKED
Incidence matrix, adjacency matrix and adjacency list of the two drawn graphs.

METHOD
Adjacency A[x][y] = 1 iff xy is an edge (arc x→y). Incidence: rows = vertices, columns = edges; undirected: 1 at both ends; directed: +1 at the tail (origin), −1 at the head.

1 Undirected graph G



G (undirected), $e_1 \ldots e_9$ as in the table

Adjacency matrix

	0	1	2	3	4	5
0	0	1	0	0	1	0
1	1	0	1	1	1	1
2	0	1	0	1	0	0
3	0	1	1	0	0	1
4	1	1	0	0	0	1
5	0	1	0	1	1	0

Adjacency list

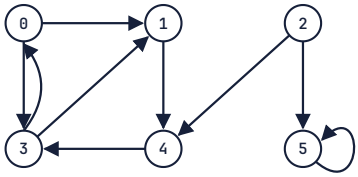
```
0 : [1, 4]
1 : [0, 2, 3, 4, 5]
2 : [1, 3]
3 : [1, 2, 5]
4 : [0, 1, 5]
5 : [1, 3, 4]
```

Incidence matrix (6 × 9):

	01	04	12	13	14	15	23	35	45
0	1	1	0	0	0	0	0	0	0
1	1	0	1	1	1	1	0	0	0
2	0	0	1	0	0	0	1	0	0
3	0	0	0	1	0	0	1	1	0
4	0	1	0	0	1	0	0	0	1
5	0	0	0	0	0	1	0	1	1

Check: each column has two 1s; row sums = degrees 2, 5, 2, 3, 3, 3 (sum 18 = 2·9).

2 Directed graph D



D (directed), $a_1 \ldots a_9$

Adjacency matrix (row = origin)

	0	1	2	3	4	5
0	0	1	0	1	0	0
1	0	0	0	0	1	0
2	0	0	0	0	1	1
3	1	1	0	0	0	0
4	0	0	0	1	0	0
5	0	0	0	0	0	1

Successor list

```
0 : [1, 3]
1 : [4]
2 : [4, 5]
3 : [0, 1]
4 : [3]
5 : [5]
```

Incidence matrix (6 × 9), arcs $a_1 \ldots a_9$:

	0→1	0→3	3→0	3→1	1→4	4→3	2→4	2→5	5→5
0	1	1	-1	0	0	0	0	0	0
1	-1	0	0	-1	1	0	0	0	0
2	0	0	0	0	0	0	1	1	0
3	0	-1	1	1	0	-1	0	0	0
4	0	0	0	0	-1	1	-1	0	0
5	0	0	0	0	0	0	0	-1	±1

Row sums of the adjacency matrix = out-degrees, column sums = in-degrees (both total 9). The loop 5→5 is both tail and head: the ±1 convention cannot encode it (it would give 0), so it is marked ±1; many courses define the incidence matrix only for graphs without loops.

ANSWER
See the matrices above. G: 6 vertices, 9 edges. D: 6 vertices, 9 arcs, including the opposite arcs 0→3 and 3→0 and the loop at 5.

Exercise 9 From matrices to graphs, and back

ASKED

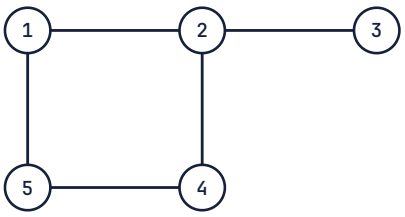
I) Draw the graph of a 5 × 5 adjacency matrix. II) Are M_1... M_4 incidence matrices of simple graphs? III) Complete the table.

METHOD

An adjacency matrix of a simple undirected graph is symmetric with a zero diagonal. An incidence matrix of a simple graph has **exactly two 1s per column** and **no two equal columns**.

1 I Graph of the adjacency matrix

The matrix is symmetric with zero diagonal, so it is a simple undirected graph on {1, ..., 5}. Reading the upper triangle: 12, 15, 23, 24, 45. It is the 4-cycle 1–2–4–5–1 with the pendant vertex 3 attached to 2.



edges 12, 15, 23, 24, 45

2 II Check the four matrices

All four have 3 rows (vertices) and 4 columns (edges). A simple graph on 3 vertices has at most C(3, 2) = 3 edges, so **none of them can be the incidence matrix of a simple graph**. The precise defect in each (highlighted):

M_1

1	1	**0**	0
1	0	**0**	1
0	1	**0**	1

column 3 is all zero: an edge with no end

M_2

0	1	0	1
1	0	**1**	1
1	1	**1**	0

columns 1 and 3 are both {2, 3}: a double edge

M_3

0	**1**	1	**1**
1	**1**	0	**1**
1	0	1	0

columns 2 and 4 are both {1, 2}: a double edge

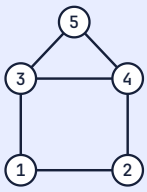
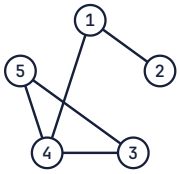
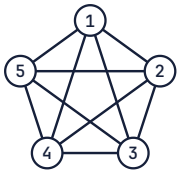
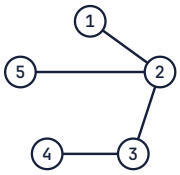
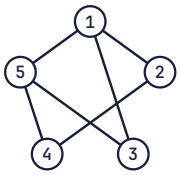
M_4

0	1	**1**	0
1	1	**1**	1
1	0	**1**	1

column 3 has three 1s: an "edge" with 3 ends (and columns 1 and 4 coincide)

3 III Completed table

The given entry of each row is the one shaded. Incidence columns are in the order of E.

Set definition	Drawing	Adjacency matrix	Incidence matrix	Adjacency list
V = {1,…,5} E = {12, 13, 24, 34, 35, 45} m = 6		0 1 1 0 0 1 0 0 1 0 1 0 0 1 1 0 1 1 0 1 0 0 1 1 0	12 13 24 34 35 45 1 1 0 0 0 0 1 0 1 0 0 0 0 1 0 1 1 0 0 0 1 1 0 1 0 0 0 0 1 1	1 : [2, 3] 2 : [1, 4] 3 : [1, 4, 5] 4 : [2, 3, 5] 5 : [3, 4]
V = {1,…,5} E = {12, 45, 34, 14, 35} m = 5		0 1 0 1 0 1 0 0 0 0 0 0 0 1 1 1 0 1 0 1 0 0 1 1 0	12 45 34 14 35 1 0 0 1 0 1 0 0 0 0 0 0 1 0 1 0 1 1 1 0 0 1 0 0 1	1 : [2, 4] 2 : [1] 3 : [4, 5] 4 : [1, 3, 5] 5 : [3, 4]
V = {1,…,5} E = {12, 13, 14, 15, 23, 24, 25, 34, 35, 45} m = 10		0 1 1 1 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0	12 13 14 15 23 24 25 34 35 45 1 1 1 1 0 0 0 0 0 0 1 0 0 0 1 1 1 0 0 0 0 1 0 0 1 0 0 1 1 0 0 0 1 0 0 1 0 1 0 1 0 0 0 1 0 0 1 0 1 1	1 : [2, 3, 4, 5] 2 : [1, 3, 4, 5] 3 : [1, 2, 4, 5] 4 : [1, 2, 3, 5] 5 : [1, 2, 3, 4]
V = {1,…,5} E = {12, 23, 25, 34} m = 4		0 1 0 0 0 1 0 1 0 1 0 1 0 1 0 0 0 1 0 0 0 1 0 0 0	12 23 25 34 1 0 0 0 1 1 1 0 0 1 0 1 0 0 0 1 0 0 1 0	1 : [2] 2 : [1, 3, 5] 3 : [2, 4] 4 : [3] 5 : [2]
V = {1,…,5} E = {12, 13, 15, 24, 35, 45} m = 6		0 1 1 0 1 1 0 0 1 0 1 0 0 0 1 0 1 0 0 1 1 0 1 1 0	12 13 15 24 35 45 1 1 1 0 0 0 1 0 0 1 0 0 0 1 0 0 1 0 0 0 0 1 0 1 0 0 1 0 1 1	1 : [2, 3, 5] 2 : [1, 4] 3 : [1, 5] 4 : [2, 5] 5 : [1, 3, 4]

- Row 1 (house): E = {12, 13, 24, 34, 35, 45}. Row 3: K_5, all 10 pairs. Row 4: the columns read 12, 23, 25, 34.
- Rows 1 and 5 are different edge sets but isomorphic graphs (both a triangle sharing an edge with a 4-cycle; for row 5: triangle 1-3-5, 4-cycle 1-2-4-5).

ANSWER

I) C_4 1-2-4-5 plus pendant 3 on 2. II) None: M_1 empty column; M_2, M_3 repeated column; M_4 a column with three 1s (and 3 vertices allow at most 3 edges anyway). III) Table above.

Exercise 10 The park benches

ASKED
1) Minimum number of colours for the benches. 2) Walk every path exactly once? 3) Visit every bench exactly once? 4) M^2, M^3 and their meaning.

METHOD
χ: lower bound ω (a clique needs distinct colours), upper bound by Brooks; then exhibit a colouring. Euler's theorem for 2. A cut vertex forbids a Hamiltonian cycle. Powers of the adjacency matrix count chains.

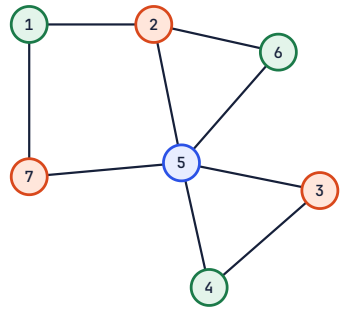
1 Read the graph

Benches 1…7, paths = 1s of M: **12, 17, 25, 26, 34, 35, 45, 56, 57** (m = 9). Degrees: d(1) = 2, d(2) = 3, d(3) = 2, d(4) = 2, d(5) = 5, d(6) = 2, d(7) = 2 (sum 18 ✓). The graph is connected.

2 1 Colouring

Lower bound: {3, 4, 5} and {2, 5, 6} are triangles, so $\omega(G) = 3$ and $\chi(G) \geq 3$ (no K_4). **Upper bound**: G is connected, neither complete nor an odd cycle, so by Brooks $\chi(G) \leq \Delta = 5$ (greedy gives $\Delta + 1 = 6$). So $3 \leq \chi \leq 5$.

The colouring below uses 3 colours and every path joins two different colours, so **$\chi(G) = 3$**.



a proper 3-colouring: blue {5}, orange {2, 3, 7}, green {1, 4, 6}

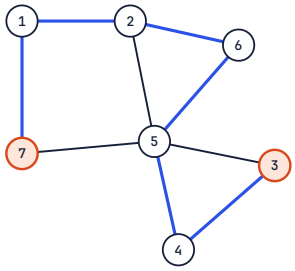
3 2 Every path exactly once

This is an Eulerian chain. G is connected and has exactly two odd vertices, 2 (degree 3) and 5 (degree 5): by Euler, an **Eulerian chain exists from 2 to 5** (but no closed tour). Example:

```
2 – 1 – 7 – 5 – 2 – 6 – 5 – 3 – 4 – 5
```

4 3 Every bench exactly once

A **Hamiltonian chain** exists: **3 – 4 – 5 – 6 – 2 – 1 – 7** (all 6 steps are paths). A Hamiltonian *cycle* does not: removing bench 5 disconnects {3, 4} from the rest, and a cycle through all vertices would survive the removal of one vertex.



Hamiltonian chain 3-4-5-6-2-1-7

5 4 **M^2 and M^3**

M^2

	1	2	3	4	5	6	7
1	2	0	0	0	2	1	0
2	0	3	1	1	1	1	2
3	0	1	2	1	1	1	1
4	0	1	1	2	1	1	1
5	2	1	1	1	5	1	0
6	1	1	1	1	1	2	1
7	0	2	1	1	0	1	2

M^3

	1	2	3	4	5	6	7
1	0	5	2	2	1	2	4
2	5	2	2	2	8	4	1
3	2	2	2	3	6	2	1
4	2	2	3	2	6	2	1
5	1	8	6	6	4	6	7
6	2	4	2	2	6	2	2
7	4	1	1	1	7	2	0

- $(M^k)_{ij}$ = number of chains of length k from i to j (sequences that may repeat vertices).
- Diagonal of M^2 = degrees (go out and come back along the same path): 2, 3, 2, 2, 5, 2, 2. Off-diagonal: number of common neighbours, e.g. $(M^2)_{15} = 2$ (via 2 and 7).
- $(M^3)_{ii}$ = 2 × number of triangles through i: trace(M^3) = 12 = 6 × 2 triangles ({2,5,6}, {3,4,5}). $(M^3)_{25} = 8$ chains of length 3 from 2 to 5.

ANSWER

1) $3 \le \chi \le 5$, and $\chi =$ **3**. 2) Yes, an Eulerian chain from 2 to 5 (not closed). 3) Yes: 3-4-5-6-2-1-7 (no Hamiltonian cycle, 5 is a cut vertex). 4) Entries count chains of length 2 and 3; diag(M^2) = degrees; trace(M^3)/6 = 2 triangles.

Exercise 11 A hypergraph

ASKED
H = (V, E), V = {a, …, h}, 8 hyperedges. 1) Incidence matrix. 2) 2-section, ω, α. 3) Stable sets vs transversals; τ(H) and α(H). 4) Link with Property 1 of the course.

METHOD
Property 1 of the course (stable ⇔ complement is a transversal), a packing of disjoint hyperedges as a lower bound for τ.

1 1 Incidence matrix

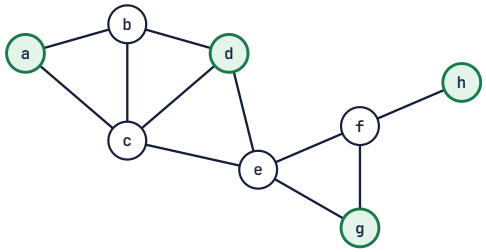
	e1	e2	e3	e4	e5	e6	e7	e8
a	1	0	0	0	0	0	0	0
b	1	1	1	0	0	0	0	0
c	1	0	1	1	1	0	0	0
d	0	1	0	1	0	0	0	0
e	0	0	0	1	1	1	0	0
f	0	0	0	0	0	1	1	1
g	0	0	0	0	0	1	1	0
h	0	0	0	0	0	0	0	1

- **Row sums** = degree of the vertex (number of hyperedges containing it): a 1, b 3, c 4, d 2, e 3, f 3, g 2, h 1.
- **Column sums** = cardinality of the hyperedge: 3, 2, 2, 3, 2, 3, 2, 2.
- For a graph every column sum is 2: M is the usual vertex–edge incidence matrix.

2 2 2-section

Replace each hyperedge by a clique on its vertices: e_1 → ab, ac, bc; e_2 → bd; e_3 → bc (already there); e_4 → cd, ce, de; e_5 → ce (already); e_6 → ef, eg, fg; e_7 → fg (already); e_8 → fh.

```
E = {ab, ac, bc, bd, cd, ce, de, ef, eg, fg, fh} (11 edges)
```



2-section; green = a maximum stable set {a, d, g, h}

- **ω = 3**: triangles abc, bcd, cde, efg. No K_4: a ∉ N(d), b ∉ N(e), and e, f, g have no common neighbour.
- **α = 4**: {a, d, g, h} is stable. Five vertices are impossible: the cliques {a,b,c}, {d,e}, {f,g}, … cover V as {a,b,c} ∪ {d,e} ∪ {f,g} ∪ {h}, four cliques, and a stable set takes at most one vertex per clique.

3 3 Stable ⇔ complement is a transversal

S contains no hyperedge ⇔ every hyperedge e has a vertex outside S ⇔ every hyperedge meets V \ S ⇔ V \ S is a transversal.

τ(H): e_2 = {b,d}, e_5 = {c,e}, e_7 = {f,g} are pairwise disjoint, and a transversal needs a different vertex in each: τ ≥ 3. T = {b, e, f} meets all eight: e_1, e_2, e_3 (b), e_4, e_5, e_6 (e), e_7, e_8 (f). So **τ(H) = 3**.

α(H) = n − τ(H) = 8 − 3 = **5**, e.g. S = V \ {b, e, f} = {a, c, d, g, h}. (The three minimum transversals are {b,e,f}, {b,c,f}, {c,d,f}; checked by program.)

4 Hyperedges of size 2

If every hyperedge has 2 vertices, H is an ordinary graph: a stable set of H is a set with no edge inside (a stable set of the graph), and a transversal meets every edge (a vertex cover). We recover **Property 1**: S stable ⇔ V \ S transversal (1.) and max|S| + min|T| = n (2.). Here the identity still holds for the hypergraph: α(H) + τ(H) = 5 + 3 = 8 = n.

Note that α(H) = 5 > α(2-section) = 4: a stable set of H may contain an edge of the 2-section (e.g. c, d), as long as it contains no whole hyperedge.

ANSWER

1) 8 × 8 matrix; rows = vertex degrees, columns = hyperedge sizes. 2) 11 edges, ω = 3, α = 4. 3) τ(H) = 3 ({b, e, f}), α(H) = 5 ({a, c, d, g, h}). 4) Property 1 for graphs; α + τ = n here too.

NOTES ON THE PROBLEM SHEET

- Ex 2.3 needs regularity of degree r ≥ 1: an edgeless graph of odd order is regular and bipartite. Ex 2.4 needs n ≥ 2.
- Ex 5.2 means "for every n ≥ 3 there is such a coloured graph", not that every G has this property.
- Ex 8: the directed graph has a loop at 5, for which the ±1 incidence convention is not defined.
- Ex 9 II: none of the four matrices is valid; with 3 vertices a simple graph has at most 3 edges, so a 3 × 4 matrix can never work.
- Ex 10.3: the answer is yes for a chain, no for a cycle.